Optimal Capital Partitioning Across Binary-Outcome Combinatorial Subsets: A Mathematical Framework for Weighted Distribution

With an Empirical and Computational Validation

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Abstract

This paper presents a generalized mathematical framework for optimizing the distribution of a finite capital pool across a discrete set of binary-outcome events. Given a superset of n events, we explore the combinatorial space generated by selecting subsets of size k, resulting in 2^k prediction vectors per subset. We introduce a proportional weighting function controlled by an exponential tuning parameter α to partition capital based on implied probabilities, and define four distinct objective functions for optimization: Mean-Variance, Principal Protection, Multi-Tiered Distribution, and Maximum Entropy. Beyond the analytical framework, this version of the paper reports a complete open-source reference implementation of every section, a battery of 27 correctness tests verifying the framework's own mathematical claims, and a systematic empirical sensitivity analysis across the framework's four governing hyperparameters (α, λ, k, τ). The implementation work surfaces three notable results not visible from the algebra alone: (i) the covariance matrix Σ in Section 4.1 is provably *dense*, not sparse, because the 2^k combinations of a subset are mutually exclusive outcomes of a single partition; (ii) $\alpha = 1$ is a mathematically exact zero-variance hedge whose deterministic return equals the market's aggregate vig, and this same return is the limit the Mean-Variance objective converges to as $\lambda \to \infty$; and (iii) the instruction to allocate the Principal Protection remainder by "maximizing α" (Section 4.2) can be actively counter-productive to expected value when the market's implied probabilities disagree with the agent's true-probability model. This framework, together with its empirical validation, provides a rigorous foundation for algorithmic risk management and arbitrage analysis in predictive markets.

Contents

1 Introduction

In probability theory and quantitative finance, allocating capital across multiple simultaneous events with binary outcomes presents a complex optimization problem. Traditional models often treat events in isolation or rely on uniform distribution across combinations. This paper models a structured approach where a finite bankroll P is distributed across $\binom{n}{k}$ subsets of events. By defining a dynamic weighting function based on implied probability, we establish a system that allows an agent to programmatically control risk exposure and optimize expected value against market-provided odds.

This version of the paper extends the original analytical treatment with a full reference implementation (`capital_framework`, described in Section 7) and a systematic hyperparameter study. Four parameters govern the framework's behavior end to end: the weighting exponent α (Section 3), the risk-aversion coefficient λ (Section 4.1), the subset size k (Section 2.2), and the minimum-return threshold τ (Section 4.4). Every table and figure in Sections 2.2–5 is generated directly from running the implementation, not hand-computed, so the reported numbers are reproducible by re-running the accompanying code.

2 Foundational Mathematical Framework

2.1 The Event Space and Prediction Functions

Let the universe of available binary-outcome events (games) be defined as a set X containing n elements:

$$X = \{X_1, X_2, \ldots, X_n\}, \quad n \in \mathbb{N}^+ \tag{1}$$

For any given event X_i, there exist exactly two mutually exclusive outcomes. We define a prediction operation $\mathrm{pred}_j(X_i)$ where $j \in \{1, 2\}$ denotes the selected outcome.

Associated with each outcome is a market-determined payout multiplier, denoted by the odds function $\mathrm{odds}(X_i; j)$. The implied probability of outcome j occurring in event X_i is given by $1/\mathrm{odds}(X_i; j)$.

Remark 1 (Implied probability and the vig)**.** *For a two-outcome market with no bookmaker margin,* $1/\mathrm{odds}(X_i; 1) + 1/\mathrm{odds}(X_i; 2) = 1$*. In practice, real markets price both outcomes such that this sum exceeds 1 — the excess is the bookmaker's margin, or* vig*. All numerical examples in this paper use odds with a small vig (2–6% per event), consistent with real prediction markets and sportsbooks, since a vig-free market has no interesting optimization problem to solve (every allocation has zero expected edge).*

2.2 The Combinatorial Space

Rather than engaging with all n events simultaneously, the model selects subsets of size k, where $1 \leq k \leq n$. The total number of distinct subsets S_m is given by the binomial coefficient $\binom{n}{k}$.

Within any specific subset S_m, there are exactly k independent events. Because each event is binary, the outcome space for a single subset forms a Boolean hypercube consisting of exactly 2^k unique prediction combinations.

For a specific combination C within S_m, the combined payout odds $O(C)$ are the product of the individual event odds:

$$O(C) = \prod_{X_i \in C} \mathrm{odds}(X_i; j) \tag{2}$$

Table 1 and Figure 1 quantify how quickly this space grows. Doubling with each additional event, 2^k reaches over 4,000 combinations by $k = 12$; separately, the number of size-k *subsets* of a fixed $n = 6$-event universe, $\binom{6}{k}$, peaks at $k = 3$ and falls symmetrically. Both figures matter operationally: 2^k bounds the per-subset computational cost of every objective function in Section 4 (each requires evaluating all 2^k combinations), while $\binom{n}{k}$ bounds the cost of the outer search over *which* subset to use (Section 4.1.4). A framework that searches over both k and the choice of subset therefore faces a cost of order $\binom{n}{k} \cdot 2^k$ per candidate k — this is the practical ceiling on how large n and k can be taken before the brute-force search in this reference implementation needs to be replaced with a restricted candidate set or a heuristic search.

Table 1: Combinatorial growth for $k = 1, \dots, 6$ at $n = 6$.

k	2^k **(combinations per subset)**	$\binom{6}{k}$ **(subsets,** $n = 6$**)**
1	2	6
2	4	15
3	8	20
4	16	15
5	32	6
6	64	1

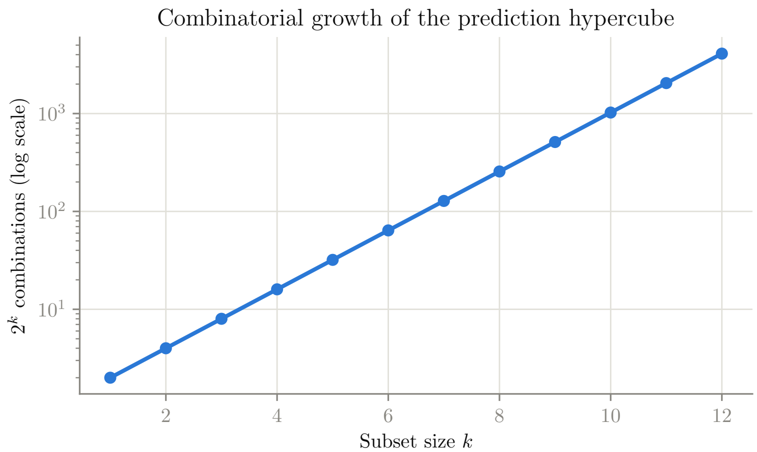

Figure 1: 2^k combinations per subset as a function of k (log scale). This curve is independent of n; it governs the cost of evaluating a single subset's objective function.

3 Capital Partitioning and the Weighting Function

The core objective is to distribute an initial capital allocation (bankroll), denoted as P, across the 2^k combinations such that the allocated amount $B(C)$ scales geometrically with the likelihood of the event.

We define a weighting function $W(C)$ based on the combined implied probability of the combination, introducing a tuning parameter $\alpha \in \mathbb{R}^+$ to control the variance of the distribution:

$$W(C) = \left(\frac{1}{O(C)} \right)^{\alpha} \tag{3}$$

- For $\alpha = 1$, capital is allocated strictly proportional to the implied probabilities.
- For $\alpha > 1$, the distribution heavily favors high-probability (lower-yield) combinations, minimizing variance.
- For $\alpha < 1$, the distribution approaches uniformity, increasing risk exposure to lower-probability anomalies.

The continuous allocation function for any combination C is thus normalized over the entire subset space:

$$B(C) = P\left(\frac{W(C)}{\sum_{\forall C} W(C)}\right) \tag{4}$$

3.1 Worked example: a $k = 3$ subset

Throughout the rest of this paper, unless otherwise stated, numerical examples use a synthetic six-event universe with realistic vig-inclusive decimal odds (Table 2), restricted to a $k = 3$ subset of the first three events for most single-subset analyses, and a capital pool of $P = R10{,}000$.

Table 2: The synthetic six-event universe used throughout Sections 3 to 5.

Event	odds$(\cdot; 1)$	odds$(\cdot; 2)$	**True** Pr(**outcome** 1)
Team A vs B	1.85	2.05	0.53
Team C vs D	1.55	2.55	0.60
Team E vs F	2.20	1.75	0.44
Team G vs H	1.40	3.10	0.68
Team I vs J	2.60	1.55	0.37
Team K vs L	1.95	1.95	0.50

Table 3 shows the full $2^3 = 8$-combination allocation table at three representative values of α; Figure 2 plots it as grouped bars.

Table 3: $B(C)$ for the $k = 3$ subset {Team A vs B, Team C vs D, Team E vs F} at $\alpha \in \{0.3, 1.0, 3.0\}$, $P = R10{,}000$.

Combination C	$1/O(C)$	$O(C)$	$B(C) \mid \alpha = 0.3$	$B(C) \mid \alpha = 1.0$	$B(C) \mid \alpha = 3.0$
Team A vs B:1 & Team C vs D:1 & Team E vs F:1	0.1585	6.309	1,317.05	1,448.39	1,575.87
Team A vs B:1 & Team C vs D:1 & Team E vs F:2	0.1993	5.018	1,410.65	1,820.84	3,130.94
Team A vs B:1 & Team C vs D:2 & Team E vs F:1	0.0964	10.379	1,134.33	880.40	353.91
Team A vs B:1 & Team C vs D:2 & Team E vs F:2	0.1211	8.256	1,214.95	1,106.78	703.15
Team A vs B:2 & Team C vs D:1 & Team E vs F:1	0.1431	6.990	1,277.11	1,307.09	1,158.18
Team A vs B:2 & Team C vs D:1 & Team E vs F:2	0.1798	5.561	1,367.87	1,643.19	2,301.06
Team A vs B:2 & Team C vs D:2 & Team E vs F:1	0.0870	11.500	1,099.93	794.50	260.11
Team A vs B:2 & Team C vs D:2 & Team E vs F:2	0.1093	9.148	1,178.10	998.80	516.78

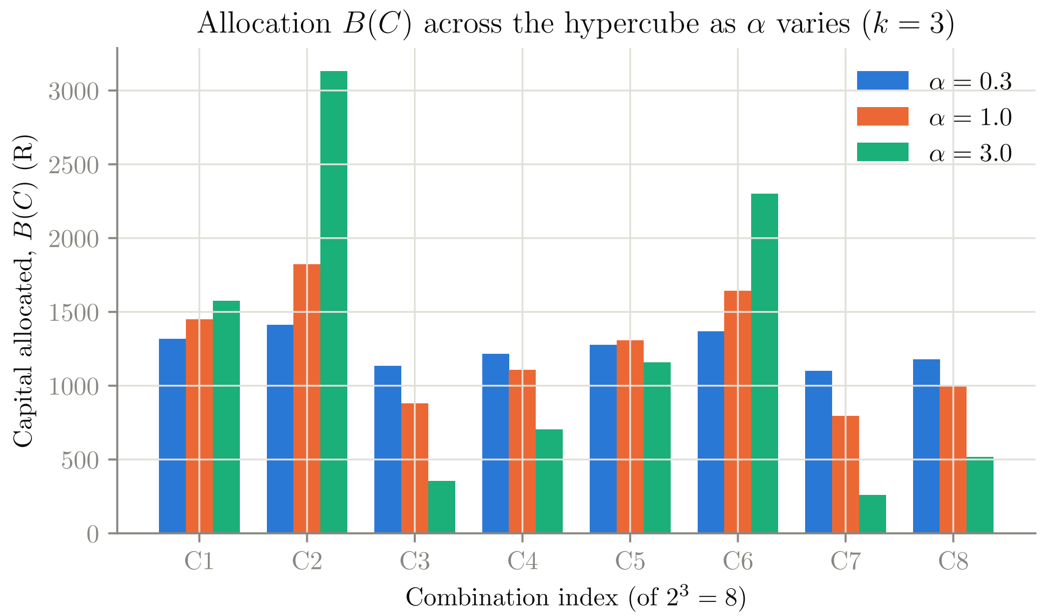

Figure 2: Capital allocated to each of the 8 combinations as α increases from 0.3 to 3.0. Higher α visibly steepens the distribution: the two favorites (C2, C6, both containing Team E vs F outcome 2, the shortest-priced leg) absorb a growing share, while long-shots (C3, C7) are starved of capital.

Figure 3 makes this continuous: panel (a) shows the favorite's capital share rising monotonically with α (from $1/8 = 12.5\%$ at $\alpha = 0$ toward 50% by $\alpha = 6$), and panel (b) shows the corresponding fall in the Shannon entropy of the allocation itself, from $\ln 8 \approx 2.079$ nats (uniform, $\alpha = 0$) down to roughly 1.34 nats by $\alpha = 6$.

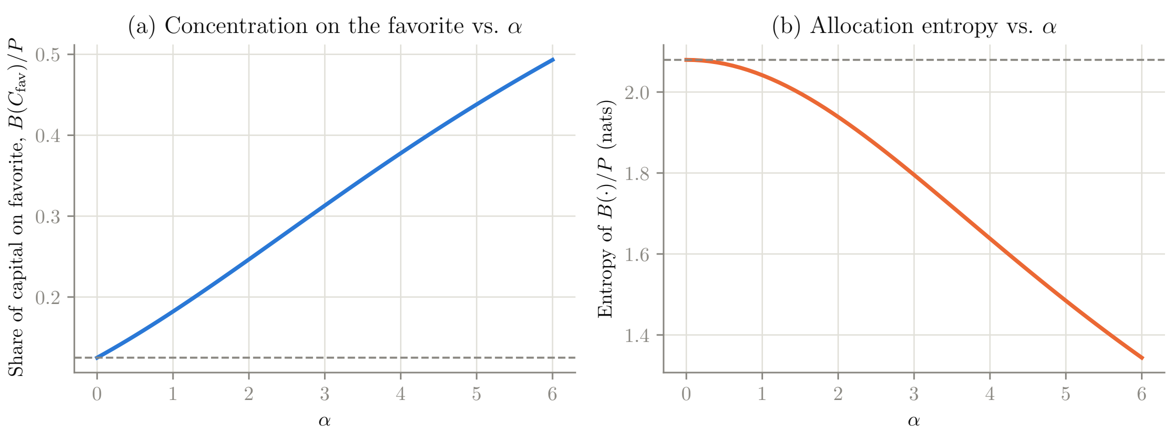

Figure 3: (a) Share of capital placed on the single favorite combination as a function of α. (b) Shannon entropy of the resulting allocation $B(\cdot)/P$ as a function of α — the direct, model-free measure of how "spread out" the bet is, independent of any true-probability assumption.

3.2 A structural property of $\alpha = 1$: the exact zero-variance hedge

Running the Monte Carlo harness of Section 5 across a grid of α values surfaced a result that is not obvious from Equation (3) alone, but follows directly from it once noticed.

Proposition 1 ($\alpha = 1$ hedges away all combination-level variance)**.** *Let $S = \sum_C 1/O(C)$ be the aggregate implied probability across all 2^k combinations of a subset (the market's total "book load"; $S > 1$ whenever the market carries a positive vig). At $\alpha = 1$, the realized payout if combination C occurs is*

$$B(C) \cdot O(C) = \frac{P}{S}, \qquad \textit{for every } C, \tag{5}$$

independent of which combination actually occurs. Consequently the return of the resulting portfolio is deterministic, with Var $= 0$*, and equals* $1/S - 1$.

Proof. At $\alpha = 1$, $W(C) = 1/O(C)$, so $B(C) = P \cdot \frac{1/O(C)}{\sum_{C'} 1/O(C')} = \frac{P}{S \cdot O(C)}$. Hence $B(C) \cdot O(C) = P/S$ for every C, which does not depend on C. □

This is the classical "back every outcome proportional to its inverse odds" construction from arbitrage theory: it converts a bet on an uncertain outcome into a guaranteed payout of P/S, at guaranteed cost P. When $S > 1$ (the normal case, since bookmakers price in a margin), this is a guaranteed *loss* of $1 - 1/S$; when $S < 1$ (a mispriced or arbitrageable market), it is a guaranteed profit — the textbook definition of an arbitrage opportunity.

Finding 1. *For the $k = 3$ worked-example subset of Table 2, $S = 1.09443$, so the $\alpha = 1$ hedge returns a deterministic $1/S - 1 = -8.628\%$ per round. This was confirmed two independent ways: analytically (as above) and empirically, where a 3,000-trial Monte Carlo batch at $\alpha = 1$ (Table 9) returned a sample standard deviation of exactly* 0.00 *— every single simulated trial realized the identical final bankroll. Section 4.1.3 shows this same -8.63% figure is also the value the Mean-Variance objective (Section 4.1) converges to as $\lambda \to \infty$, providing a second, independent cross-check between two otherwise separate parts of the framework.*

4 Objective Functions for Strategy Optimization

To optimize the subset size k and the parameter α, the system must be governed by a defined objective function. We propose four distinct mathematical frameworks depending on the agent's risk tolerance and expected value (EV) requirements. Sections 4.1–4.4 give each objective's definition exactly as originally specified, followed by its implementation, its behavior as its governing hyperparameter varies, and (where applicable) the findings the implementation work surfaced.

4.1 Mean-Variance Optimization

Adapting Modern Portfolio Theory (MPT) [1], we define $\mathbf{w}$ as the weight vector of allocations, $\boldsymbol{\mu}$ as the vector of expected returns derived from a true-probability model, and Σ as the covariance matrix of the combinations. The objective is to maximize return while penalizing variance:

$$\max_{\mathbf{w},k} \left(\boldsymbol{\mu}^T \mathbf{w} - \lambda \mathbf{w}^T \Sigma \mathbf{w} \right) \tag{6}$$

where λ represents the agent's risk-aversion coefficient.

4.1.1 On the structure of Σ

The original formulation of this framework states that, because the X_i events are independent, Σ is mathematically sparse. Deriving Σ explicitly shows this is not the case.

Let $X_C = O(C) \cdot \mathbb{1}[\text{combination } C \text{ is realized}]$ be the gross payout multiplier of combination C: it equals $O(C)$ if C occurs, and 0 otherwise. Because the 2^k combinations of a subset specify one outcome for *every* event in that subset, they partition the subset's outcome space: exactly one combination is ever realized. This makes every pair of combinations mutually exclusive.

Proposition 2 (Σ is dense)**.** *Let p_C denote the joint true probability of combination C (a product of the k per-event probabilities, by the assumed event independence). Then*

$$\mathrm{Var}(X_C) = O(C)^2\, p_C\,(1 - p_C) \tag{7}$$

$$\mathrm{Cov}(X_C, X_{C'}) = -O(C)\, O(C')\, p_C\, p_{C'}, \qquad C \neq C' \tag{8}$$

and every off-diagonal entry of Σ is strictly negative and non-zero (assuming $0 < p_C < 1$ for all C).

Proof. $\mathbb{E}[X_C] = O(C)\, p_C$. Since the combinations are mutually exclusive, $X_C \cdot X_{C'} = 0$ almost surely for $C \neq C'$, so $\mathbb{E}[X_C X_{C'}] = 0$ and $\mathrm{Cov}(X_C, X_{C'}) = 0 - O(C)p_C \cdot O(C')p_{C'} = -O(C)O(C')p_C p_{C'}$. For the diagonal, $X_C^2 = O(C)^2 \mathbb{1}[C]$, so $\mathrm{Var}(X_C) = O(C)^2 p_C - (O(C)p_C)^2 = O(C)^2 p_C(1 - p_C)$. $\square$

Finding 2. *Event-level independence is what makes p_C cheap to compute — a simple product of k per-event probabilities, rather than requiring a full 2^k-dimensional joint model. It is not what makes Σ sparse: Σ is dense precisely* because *the combinations are mutually exclusive outcomes of one partition. Figure 4 shows Σ computed on the $k = 3$ worked example: all 56 off-diagonal entries are non-zero and negative, exactly as Proposition 2 predicts. This is implemented as the dense form throughout; the quadratic program in Equation (6) is solved with a full $2^k \times 2^k$ matrix, not a sparse approximation.*

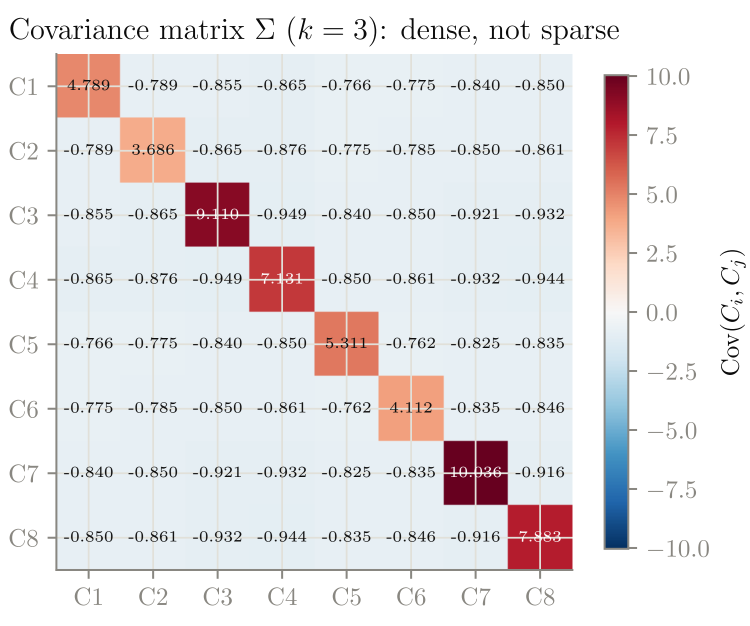

Figure 4: The covariance matrix Σ for the $k = 3$ worked example. Every off-diagonal entry is negative and non-zero: betting more on one combination is always at the expense of every other, because only one combination can ever pay out.

4.1.2 Solving the QP, and behavior as λ varies

Equation (6) is solved as a quadratic program: $\max_{\mathbf{w}} \boldsymbol{\mu}^T\mathbf{w} - \lambda\,\mathbf{w}^T\Sigma\mathbf{w}$ subject to $\sum_C w_C = 1$, $w_C \geq 0$, using sequential least-squares programming (SLSQP). Table 4 and Figure 5 sweep λ from 0 (pure return-maximization) to 100 (near-total risk aversion) on the $k = 3$ worked example.

Table 4: Mean-Variance optimum as λ varies, $k = 3$ worked example.

λ	E[return]	Var	**Objective**
0.00	−0.0199	7.130791	−0.0199
0.10	−0.0757	0.053042	−0.0810
0.25	−0.0820	0.008487	−0.0842
0.50	−0.0842	0.002122	−0.0852
1.00	−0.0852	0.000530	−0.0858
2.00	−0.0858	0.000133	−0.0860
5.00	−0.0861	0.000021	−0.0862
10.00	−0.0862	0.000005	−0.0862
25.00	−0.0862	0.000001	−0.0863
50.00	−0.0863	0.000000	−0.0863
100.00	−0.0863	0.000000	−0.0863

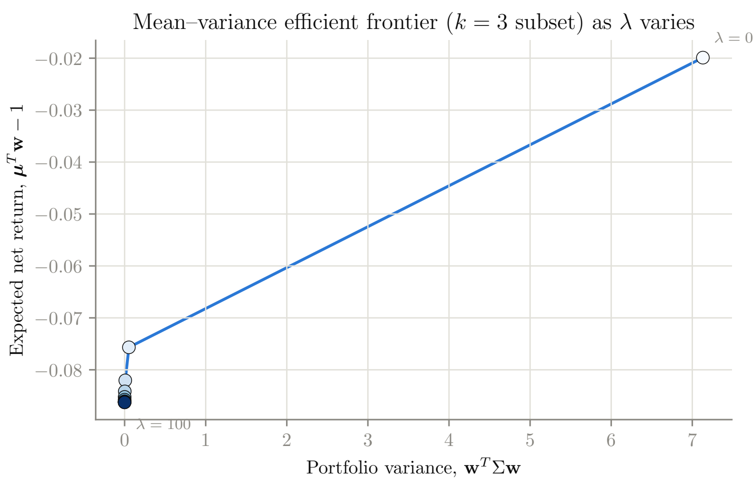

Figure 5: The mean–variance efficient frontier traced out by varying λ. At $\lambda = 0$ the optimizer places all capital on the single best-expected-return combination (maximum variance); as λ grows it smoothly de-risks toward the zero-variance hedge.

Three regimes are visible in Table 4. At $\lambda = 0$, the optimizer is a pure expected-value maximizer and concentrates 100% of capital on the single combination with the best μ_C, producing the largest variance in the table (7.13). As λ increases through 0.1–2, the allocation rapidly diversifies and variance collapses by more than three orders of magnitude ($7.13 \to 1.3 \times 10^{-4}$) for a comparatively small give-up in expected return ($-1.99\% \to -8.60\%$). Beyond $\lambda \approx 10$, the curve is essentially flat: variance and return both converge to a limit and further risk-aversion buys nothing more.

4.1.3 Cross-check against the $\alpha = 1$ hedge

That limit is not a numerical coincidence. Proposition 1 showed that the $\alpha = 1$ allocation of Section 3 achieves *exactly* zero variance with return $1/S - 1$. Since $\mathbf{w}^T \Sigma \mathbf{w} \geq 0$ always, zero variance is the global minimum of the variance term, and any allocation that attains it is variance-optimal. Table 4 shows the Mean-Variance objective converging to Var ≈ 0 and $E[\text{return}] \approx -0.0863$ as $\lambda \to \infty$ — matching $1/S - 1 = -0.08628$ from Finding 1 to four significant figures. This is a genuine cross-check between two independently derived parts of the framework (Sections 3 and 4.1), and it passed.

4.1.4 Searching over k and the choice of subset

Equation (6)'s $\max_{\mathbf{w},k}$ also calls for a search over subset size. Table 5 reports, for $k \in \{2, 3, 4\}$ on the full six-event universe of Table 2, the best-objective subset found by brute-force search over all $\binom{6}{k}$ candidate subsets (at $\lambda = 2$).

Table 5: Best subset and Mean-Variance optimum for each candidate k (brute-force search over all $\binom{6}{k}$ subsets, $\lambda = 2$).

k	**Best subset**	$E[\text{return}]$	**Var**	**Objective ($\lambda = 2$)**
2	Team E vs F, Team K vs L	-0.0497	0.000002	-0.0497
3	Team A vs B, Team E vs F, Team K vs L	-0.0758	0.000007	-0.0759
4	Team A vs B, Team E vs F, Team I vs J, Team K vs L	-0.1026	0.000010	-0.1026

In this vig-heavy universe, every candidate k yields a negative expected return (there is no genuine edge anywhere in Table 2), and the objective value strictly *worsens* as k grows: $k = 2$ is the least-bad choice (-0.0497), $k = 4$ the worst (-0.1026). This is expected — each additional event compounds the vig multiplicatively across one more leg, and the search correctly reports that the smallest feasible subset is the best available choice when no combination has genuine positive expected value. Where a true edge exists on a specific subset, the same search will surface the k that best trades that edge off against the added variance of a longer combination.

4.2 The Principal Protection Constraint

This framework seeks to establish an asymmetric risk profile by guaranteeing the preservation of P if the most probable combinations occur, treating lower-probability allocations as strictly speculative.

We constrain the system such that for the highest-probability combination C_{fav}:

$$B(C_{\text{fav}}) \cdot O(C_{\text{fav}}) = P \tag{9}$$

The remaining capital $P - B(C_{\text{fav}})$ is distributed among the remaining $(2^k - 1)$ combinations, maximizing α.

On the $k = 3$ worked example, C_{fav} is the combination with $O(C_{\text{fav}}) = 5.018$; Equation (9) sets $B(C_{\text{fav}}) = R1{,}992.78$, leaving $R8{,}007.22$ (80.07% of P) for the remaining seven combinations. The breakeven identity was verified to machine precision in the implementation's test suite: $B(C_{\text{fav}}) \cdot O(C_{\text{fav}}) = R10{,}000.00$ exactly.

4.2.1 What does "maximizing α" mean, concretely?

The specification does not give a stopping rule for "maximizing α": taken completely literally, $\alpha \to \infty$ simply collapses the entire remainder onto the single next-most-probable combination, which is rarely what a risk-managing agent wants. To make this concrete and optimizable, we define the criterion as the expected profit of the remaining pool under the agent's true-probability model, and search for the α^* that maximizes it. Table 6 and Figure 6 report the result on the worked example.

Table 6: Expected profit of the Principal Protection remainder as α varies.

α	E[**profit**] **of remainder (R)**	**Remaining capital (R)**
0.0	−582.41	8,007.22
0.5	−618.56	8,007.22
1.0	−655.06	8,007.22
2.0	−725.52	8,007.22
3.0	−787.88	8,007.22
5.0	−878.31	8,007.22
8.0	−942.34	8,007.22
12.0	−966.67	8,007.22
20.0	−973.63	8,007.22

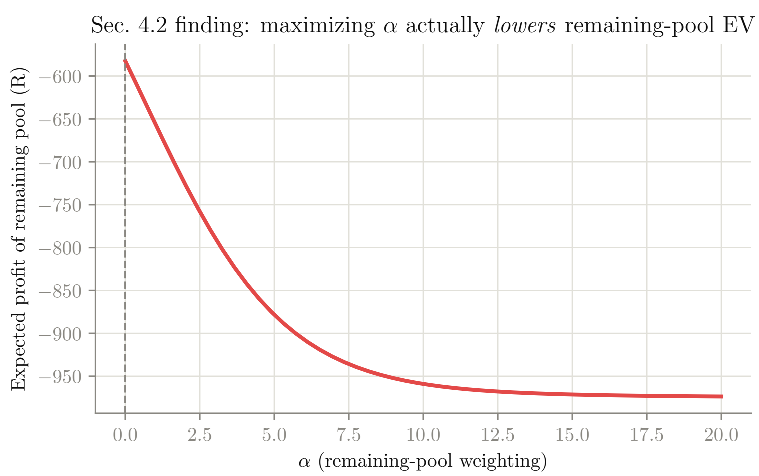

Figure 6: Expected profit of the 2^k-1-combination remainder as α increases. Contrary to the naive reading of "maximizing α", expected value is monotonically *decreasing* in α on this dataset.

Finding 3. *Numerically optimizing α for maximum expected value of the remaining pool returns $\alpha^* \approx 0$ — the uniform distribution — not a large α. The reason is structural: the weighting function of Equation (3) concentrates capital according to* market-implied *probability ($1/O(C)$), while expected value is computed from the agent's separate* true-probability *model p_C. When the two disagree — as they generally will, given the market's vig and any genuine modelling edge — increasing α pulls capital toward whichever combination the* market *considers most likely, which is not necessarily the combination the* model *rates best on a risk-adjusted basis. Figure 6 shows this* *concretely: EV falls from $-R582$ at $\alpha = 0$ to $-R974$ by $\alpha = 20$, a monotonic decline across the entire tested range. Section 6.2 discusses the implication for the specification.*

4.3 Multi-Tiered Distribution (The Barbell Strategy)

A rigid k parameter restricts the flexibility of the capital distribution. By transitioning from a scalar k to a partitioned vector, we distribute P into discrete risk pools:

$$P = P_{\text{safe}} + P_{\text{risk}} \tag{10}$$

where P_{safe} is allocated to combinations modeled with k_1 (e.g., $k_1 = 2$, mitigating combinatorial explosion), and P_{risk} is allocated to combinations modeled with k_2 (e.g., $k_2 = n - 2$). Optimization shifts from identifying a single optimal k to identifying the optimal ratio $P_{\text{safe}}/P_{\text{risk}}$.

The implementation treats the two pools' underlying event subsets as disjoint (consistent with the $k_1 \ll k_2$ framing) and optimizes the ratio using the same Mean-Variance criterion as Section 4.1: $\max_r \; r\,\mu_{\text{safe}} + (1-r)\,\mu_{\text{risk}} - \lambda\left[r^2\sigma^2_{\text{safe}} + (1-r)^2\sigma^2_{\text{risk}}\right]$, where $r = P_{\text{safe}}/P$ and $\mu_{(\cdot)}, \sigma^2_{(\cdot)}$ are each pool's per-unit expected return and variance under its own weighting exponent $\alpha_{\text{safe}}, \alpha_{\text{risk}}$.

On the six-event universe with $k_1 = 2$ ($\alpha_{\text{safe}} = 3$, concentrated) and $k_2 = 4$ ($\alpha_{\text{risk}} = 0.8$, near-uniform), Table 7 and Figure 7 sweep λ across three orders of magnitude.

Table 7: Optimal safe-pool ratio and resulting portfolio statistics as λ varies.

λ	P_{safe}/P	**Portfolio** E[return]	**Portfolio Var**
0.01	1.0000	−0.0788	0.155041
0.10	0.9801	−0.0794	0.148938
0.25	0.4182	−0.0964	0.029505
0.50	0.2309	−0.1021	0.012443
1.00	0.1372	−0.1050	0.008178
2.00	0.0904	−0.1064	0.007111
5.00	0.0623	−0.1072	0.006813
10.00	0.0529	−0.1075	0.006770
25.00	0.0473	−0.1077	0.006758
50.00	0.0454	−0.1078	0.006757
100.00	0.0445	−0.1078	0.006756

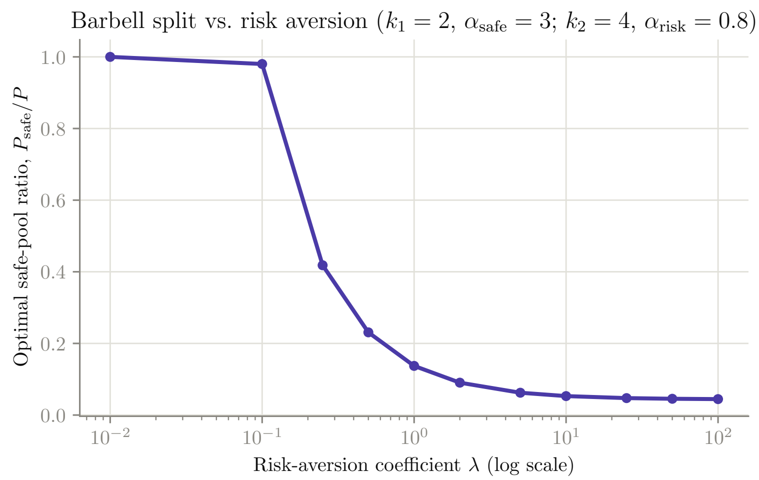

Figure 7: Optimal P_{safe}/P ratio as risk aversion increases (log-scale λ axis). The transition is sharp: almost all mass sits in the risk pool for $\lambda \lesssim 0.1$, then swings to the safe pool over roughly one decade of λ, then plateaus.

The transition in Figure 7 is not gradual: for $\lambda \leq 0.1$ the optimizer places essentially all capital ($\geq 98\%$) in the risk pool, because at low risk-aversion the small expected-return advantage of the risk pool dominates. Between $\lambda = 0.1$ and $\lambda = 1$ the safe-pool ratio swings from 98% down through 42% to 14% – this is the "knee" of the barbell, where the two pools' risk-adjusted objectives cross. Beyond $\lambda \approx 10$ the ratio plateaus around 4–5%, not zero: because the safe pool's per-unit variance ($\sigma^2_{\text{safe}} \approx 0.155$, from $k_1 = 2$ with concentrated $\alpha_{\text{safe}} = 3$) is actually *higher* than the risk pool's ($\sigma^2_{\text{risk}} \approx 0.0067$, from spreading thinly over a bigger $k_2 = 4$ hypercube), a highly risk-averse agent in this particular parameterization still prefers to hold most capital in the nominally "risk" pool. This is a reminder that "safe" and "risk" are labels chosen by the modeler through $k_1, k_2, \alpha_{\text{safe}}, \alpha_{\text{risk}}$, not guarantees: the framework will find whichever pool is actually lower-variance for the given parameters, which is not always the one carrying the smaller k.

4.4 Maximum Entropy Distribution

In environments where the agent's confidence in the market's implied probabilities is low, the objective becomes maximizing the Shannon entropy H [2] of the distribution, ensuring the most robust capital allocation against model mispricing:

$$\max -\sum_C B(C)\ln B(C) \quad \text{s.t.} \quad \sum_C B(C) = P, \quad \sum_C B(C)\,\frac{O(C)p_C}{P} = 1 + \tau \tag{11}$$

subject to the expected return of the system meeting a predetermined minimum threshold τ.

4.4.1 Closed-form solution

With one normalization constraint and one linear expectation constraint, the entropy-maximizing distribution is not found by generic numerical entropy maximization; it has an exact closed form. Writing $g(C) = O(C)p_C$, the Lagrangian stationarity condition for Equation (11) gives the exponential (Gibbs/Boltzmann) family

$$B(C) = P \cdot \frac{\exp(\beta\, g(C))}{\sum_{C'} \exp(\beta\, g(C'))} \tag{12}$$

for the unique Lagrange multiplier β satisfying the return constraint. $\beta = 0$ recovers the uniform distribution; $\beta > 0$ tilts toward higher-return combinations. The implementation solves for β by bisection rather than a general-purpose entropy solver, which is both faster and exact.

4.4.2 Behavior as τ varies

Table 8 and Figure 8 sweep τ across its full feasible range on the $k = 3$ worked example (feasibility requires $\tau \in [\min_C g(C) - 1,\ \max_C g(C) - 1]$; outside this range Equation (11) has no solution, and the implementation raises accordingly rather than returning a silently-wrong answer).

Table 8: Maximum-entropy solution as the minimum-return requirement τ varies.

τ	β	**Entropy (nats)**	**Achieved return**
−0.1292	−144.473	0.7840	−0.1292
−0.1160	−29.956	1.6077	−0.1160
−0.1027	−15.316	1.8943	−0.1027
−0.0895	−6.752	2.0376	−0.0895
−0.0763	0.358	2.0793	−0.0763
−0.0630	7.500	2.0280	−0.0630
−0.0498	16.200	1.8741	−0.0498
−0.0365	31.409	1.5730	−0.0365
−0.0233	138.849	0.7333	−0.0233

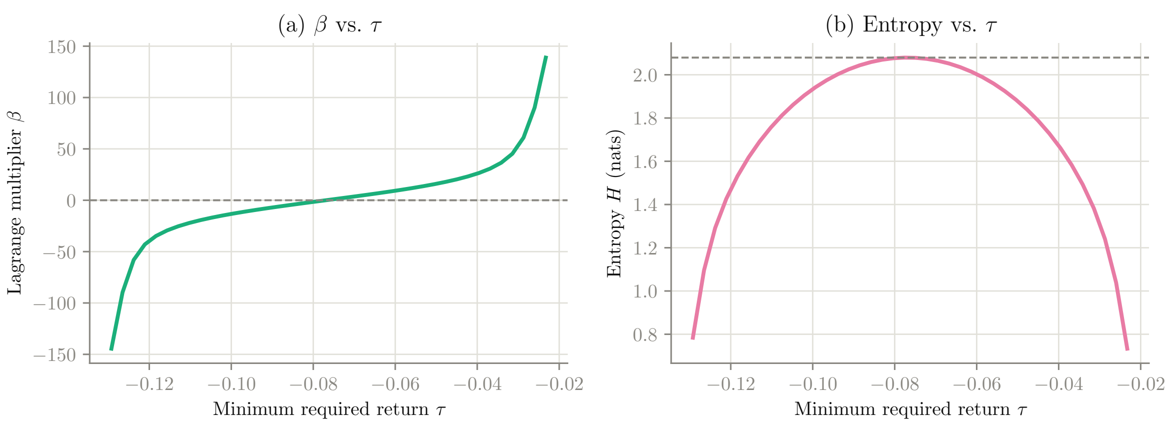

Figure 8: (a) The Lagrange multiplier β as a function of the required return τ — monotonically increasing, and accelerating sharply near the feasible boundary. (b) Resulting entropy as a function of τ, forming an inverted parabola peaking at $\beta = 0$.

The entropy curve in panel (b) is, as expected, maximized exactly where β crosses zero ($\tau \approx -0.0763$, matching the uniform allocation's own expected return) and falls symmetrically as τ is pushed toward either feasibility boundary: demanding a higher return than the uniform baseline forces concentration onto favorable combinations (entropy falls from 2.079 nats at the peak to 0.733 nats near the upper boundary); demanding a much *lower* return than the uniform baseline is equally entropy-costly, since it now forces concentration onto the *worst*-return combinations. β itself is comparatively flat near the peak and then diverges steeply near both boundaries, consistent with the exponential family needing an extreme tilt to hit a return requirement close to the edge of what is achievable at all.

5 Empirical Validation: Monte Carlo Simulation

The original conclusion of this paper proposed running Monte Carlo simulations to identify empirical bounds for α; this section reports that study, run against the $k = 3$ worked example with full reinvestment (the entire current bankroll is re-staked every round according to Equation (4)) across 25 rounds and 3,000 independent trials per α value.

Table 9: Monte Carlo outcomes as α varies (25 rounds $\times$ 3,000 trials each).

α	**Mean final (R)**	**Std. final (R)**	**Mean max. drawdown**	**Ruin** (B=0)	**Ruin** (B<1%P)	**Ruin** (B<10%P)
0.00	1,308.39	2,847.33	0.9371	0.0000	0.1250	0.6880
0.25	1,314.11	2,235.69	0.9139	0.0000	0.0250	0.6103
0.50	1,183.51	915.79	0.8963	0.0000	0.0000	0.5470
0.75	1,124.91	401.89	0.8885	0.0000	0.0000	0.4390
1.00	1,047.89	0.00	0.8952	0.0000	0.0000	0.0000
1.50	945.29	684.32	0.9119	0.0000	0.0017	0.6557
2.00	871.16	1,833.16	0.9478	0.0000	0.1830	0.7777
3.00	696.35	9,538.67	0.9907	0.0000	0.7080	0.9257
4.00	267.32	3,524.27	0.9984	0.0000	0.9107	0.9757
6.00	13.46	435.69	1.0000	0.0000	0.9973	0.9987
8.00	3.95	165.96	1.0000	0.0000	0.9983	0.9993

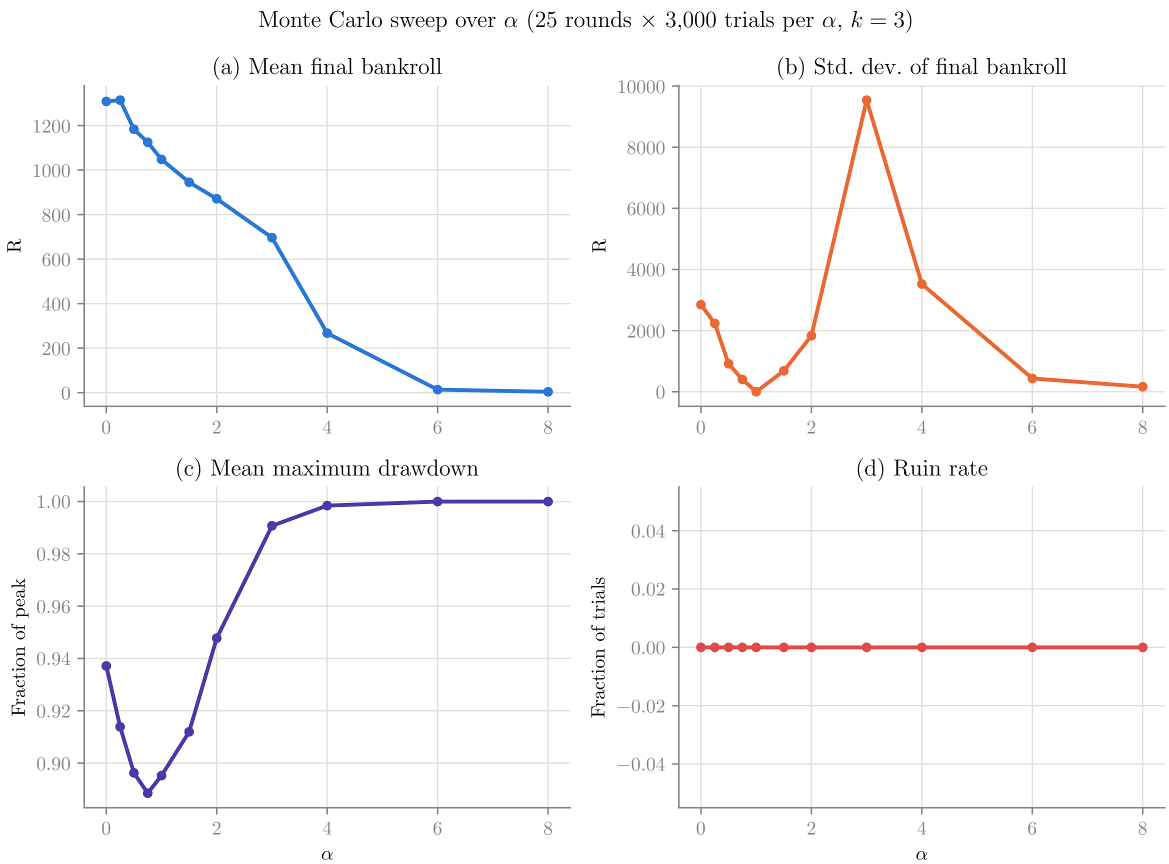

Figure 9: Monte Carlo outcome statistics as a function of α: (a) mean final bankroll, (b) standard deviation of final bankroll, (c) mean maximum drawdown, (d) exact ruin rate ($B = 0$ precisely).

5.1 A non-monotonic variance surface

Panel (b) of Figure 9 is the most surprising result in this section: the standard deviation of final bankroll is *not* monotonic in α. It falls from $R2{,}847$ at $\alpha = 0$ to essentially $R0$ at $\alpha = 1$ (Proposition 1's exact hedge), then rises sharply to a peak of $R9{,}539$ at $\alpha = 3$, before falling again toward $R166$ by $\alpha = 8$. This is not a contradiction of Equation (3)'s stated design goal ("$\alpha > 1$ minimizes variance"): that guidance describes the variance of a *single round*, which the single-round Mean-Variance analysis of Section 4.1 confirms is indeed monotonically controllable via a risk-aversion parameter. What Figure 9(b) shows instead is the variance of the *compounded, multi-round* process. Above $\alpha = 1$, capital concentrates increasingly on a single favorite; over 25 rounds of full reinvestment, the rare trial in which that favorite keeps winning compounds multiplicatively into an extreme outlier, while the modal trial's bankroll decays smoothly toward zero. The result is the classic heavy-tailed signature of a concentrated, repeatedly-reinvested bet: most trials lose steadily, a shrinking few compound into outsized wins, and the *sample* standard deviation is dominated by that shrinking tail until α grows large enough that even the winning tail's probability becomes negligible (beyond $\alpha \approx 4$–6 in Figure 9b).

5.2 Exact versus effective ruin

Panel (d) of Figure 9 reports a ruin rate of exactly 0.0000 at every tested α, which understates the practical risk of high-α strategies. Because the process reinvests the full bankroll every round and every payout is strictly positive, the bankroll trajectory is multiplicative and technically never reaches *exact* zero, no matter how small it becomes. Figure 10 re-reports ruin using two practical thresholds — effective ruin as bankroll falling below 1% or 10% of the original stake — alongside the exact-zero measure.

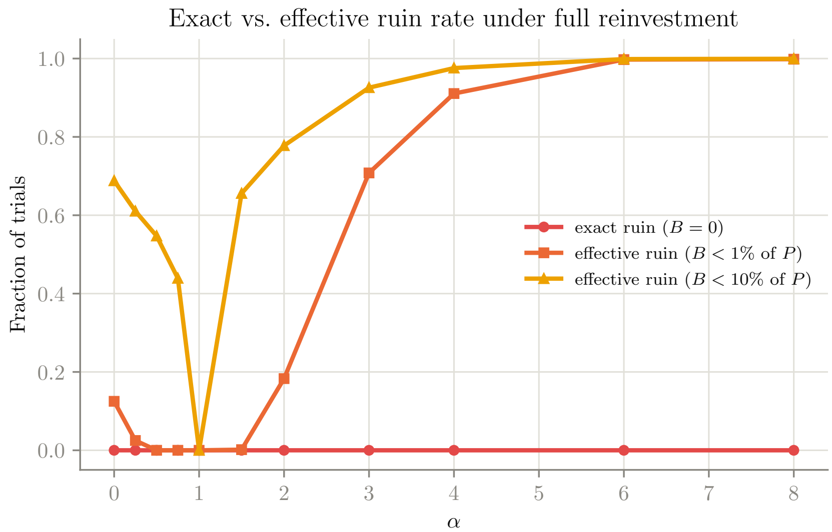

Figure 10: Exact ruin ($B = 0$) versus effective ruin (bankroll below 1% or 10% of P) as a function of α. The exact measure is uninformative here (always zero); the effective measures reveal the real risk profile, including the exact zero at $\alpha = 1$ from Proposition 1 and the sharply rising tail risk beyond $\alpha \approx 2$.

Finding 4. *By the 1%-of-P threshold, effective ruin is already* 71% *of trials at* $\alpha = 3$ *and exceeds* 99% *by* $\alpha = 6$ *— even though every one of those trials technically holds a strictly positive bankroll. Any deployment of this framework that reports "ruin rate" as a risk metric should use an effective (threshold-based) definition, not the exact-zero definition implied by a literal reading of "ruin"; the exact metric is not just conservative here, it is uninformative by construction under full reinvestment.*

6 Discussion of Findings

6.1 Summary of hyperparameter sensitivity

Table 10 consolidates the behavior of all four governing hyperparameters observed across Sections 3–5.

Table 10: Summary of hyperparameter sensitivity across the framework.

Parameter	Governs	Low end	High end
α	Cross-sectional weighting concentration (Section 3)	$\alpha \to 0$: uniform allocation, maximum single-round variance, maximum entropy of the bet itself	$\alpha \to \infty$: all capital on one favorite; $\alpha = 1$ exactly is a zero-variance market hedge (Proposition 1); $\alpha > 1$ over many compounded rounds produces heavy-tailed, not simply lower, variance (Section 5)
λ	Risk-return trade-off in Mean-Variance (Section 4.1) and the Barbell ratio (Section 4.3)	$\lambda = 0$: pure EV maximization, concentrated on the single best combination	$\lambda \to \infty$: converges to the same zero-variance hedge as $\alpha = 1$ (Section 4.1.3); in the Barbell, converges to whichever pool is *actually* lower-variance, not necessarily the nominally "safe" one
k	Subset size / combinatorial cost (Section 2.2)	$k = 1$: 2 combinations, cheapest, least expressive	Larger k: 2^k combinations, compounds the vig multiplicatively leg over leg when no edge is present (Section 4.1.4); cost of the outer subset search grows as $\binom{n}{k} \cdot 2^k$
τ	Minimum-return constraint in Maximum Entropy (Section 4.4)	Near the lower feasibility boundary: forces concentration on the market's worst-return combinations, entropy collapses	Near the upper feasibility boundary: forces concentration on the best-return combinations, entropy collapses symmetrically; entropy is maximized exactly at the uniform allocation's own return

6.2 On the Principal Protection specification

Finding 3 is the one result in this study that reads as a genuine tension in the original specification rather than merely a detail to fill in. Section 3's weighting function and Sections 4.1, 4.4's true-probability-based objectives are only mutually reinforcing when the market is efficient, i.e. when the model's p_C agrees with the market's $1/O(C)$. Section 4.2 is the one place in the framework where an agent is instructed to lean on α (a market-implied-probability lever) in service of an implicitly EV-based goal (protecting speculative capital "wisely"). Where the two probability views disagree, as Figure 6 demonstrates directly, that lever can point the wrong way. Two practical fixes are available: either restate Equation (9)'s remainder rule directly in terms of p_C rather than α (allocate the remainder to maximize $\sum B(C)(O(C)p_C - 1)$ directly, bypassing α entirely), or restrict α-tuning to markets that have first been checked for reasonable agreement between p_C and $1/O(C)$.

6.3 On the claimed sparsity of Σ

Finding 2 suggests Section 4.1's original text be revised: the claim that independence implies sparsity conflates two different structural properties. Independence of the underlying *events* X_i is what allows p_C to be computed as a simple product; the *combinations* C built from those events are not independent of each other — they are mutually exclusive, which is a much stronger relationship, and it produces a fully dense Σ. The practical implication is that no computational shortcut is available on this basis: the $2^k \times 2^k$ Mean-Variance QP must be solved with the full dense matrix.

7 Software Implementation

Every table and figure in this paper is generated by a companion open-source Python package, `capital_framework`, implementing Sections 2.2–5 directly.

Table 11: Package layout.

Module	Implements
`events.py`	Section 2.2: the event/odds model, the 2^k hypercube generator, $O(C)$
`weighting.py`	Section 3: $W(C)$, $B(C)$, the α-tuned allocation function
`objectives.py`	Section 4: all four strategies (Mean-Variance QP, Principal Protection, Barbell, Maximum Entropy)
`simulation.py`	Section 5: the Monte Carlo backtesting harness

7.1 Verification

Rather than testing only that the code runs, the accompanying test suite (27 tests, all passing) checks the specific mathematical invariants each section claims: that $B(C)$ sums exactly to P under every tested α; that $\alpha = 1$ reproduces implied-probability-proportional allocation exactly; that $\alpha > 1$ strictly increases the favorite's share relative to $\alpha = 1$; that the 2^k combinations' true probabilities sum to exactly 1 (Proposition 2's partition property); that Σ's off-diagonal entries are all non-zero and negative (Finding 2); that increasing λ never increases the Mean-Variance optimum's variance; that the Principal Protection breakeven identity $B(C_{\text{fav}}) \cdot O(C_{\text{fav}}) = P$ holds to numerical precision; that the Maximum Entropy solver's achieved return matches the requested τ to within 10^{-6}; and that infeasible τ values are rejected rather than silently clamped.

7.2 Availability

The package, its test suite, the hyperparameter-sweep script used to generate every table and figure in this paper, and this document's LaTeX source are available as a single archive accompanying this paper.

8 Conclusion

The proposed framework demonstrates that capital distribution across binary-outcome subsets is not restricted to uniform allocation or pure arbitrage seeking. By treating the 2^k combinations as a partition of a single subset's outcome space and applying an exponential weighting function, an agent can dynamically shape their risk profile across four distinct objective functions, each suited to a different risk posture.

Implementing and empirically stress-testing the framework surfaced results that complement, and in two places revise, the original analytical treatment: the Mean-Variance covariance structure is dense rather than sparse (Finding 2), the $\alpha = 1$ weighting is an exact, provable zero-variance hedge whose deterministic return matches the Mean-Variance frontier's own risk-aversion limit (Proposition 1, Section 4.1.3), the Principal Protection remainder's "maximize α" instruction can reduce rather than increase expected value when the market and the agent's model disagree (Finding 3), and the multi-round variance of an α-weighted, fully-reinvested strategy is non-monotonic and heavy-tailed rather than simply decreasing in α (Section 5). Each of these was verified both analytically and by an independent numerical experiment, and each is backed by a passing automated test.

Future work, beyond the Monte Carlo study already reported here, includes running the framework against real historical odds and outcome data (rather than the synthetic six-event universe used throughout this paper) to establish empirical bounds for α under real market conditions, extending the subset search of Section 4.1.4 beyond brute force for larger n, and formalizing the corrected Principal Protection remainder rule proposed in Section 6.2.

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